



TEST CODE **02107032**

FORM TP 2009148

MAY/JUNE 2009

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

ADVANCED PROFICIENCY EXAMINATION

BIOLOGY

UNIT 1 – PAPER 03/2

ALTERNATIVE TO INTERNAL ASSESSMENT

2 hours

**Candidates are advised to use the first 15 minutes for
reading through this paper carefully.**

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of THREE questions. Answer ALL questions.
2. The use of silent non-programmable calculators is allowed.

1. Amylase is an enzyme, which catalyses the hydrolysis of starch into a reducing sugar. It is commonly found in human saliva and germinating seeds.

You are required to carry out a simple investigation into the effect of substrate concentrations on the rate of reaction of amylase. Carefully read the instructions that follow before starting.

- (a) You are provided with a stock solution of starch.
Using the distilled water provided prepare a series of starch solutions at different concentrations as follows:

(i) Label 8 test tubes A, B, C, D, A1, B1, C1, D1

(ii) To EACH test tube add the following:

A: 5 cm³ stock starch solution

B: 4 cm³ stock starch solution and 1 cm³ distilled water

C: 3 cm³ stock starch solution and 2 cm³ distilled water

D: 2 cm³ stock starch solution and 3 cm³ distilled water

A1, B1, C1, D1: 5 cm³ distilled water

- (b) Place all the test tubes in a test tube rack in the water bath (37 °C) provided and leave for 5 minutes to allow the temperature of the solutions to reach the temperature of the water bath.

- (c) Remove test tubes A and A1 from the water bath. To each tube quickly add 0.5 cm³ of the amylase enzyme solution provided. Swirl the contents of the test tubes and rapidly put them back into the rack in the water bath.

- (d) START THE STOP CLOCK. At intervals of 1 minute remove 1 drop of the reaction mixture from each test tube and place in separate wells of the marble spot plate provided. Immediately add a drop of Benedict's solution. Record the starch concentration and colour changes in Table 1 on page 3. Repeat this for a period of 5 minutes.

- (e) Carry out the same procedure outlined in Steps (c) and (d) for the remaining pairs of solutions as follows:

B and B1

C and C1

D and D1

[7 marks]

- (f) Write an appropriate title for Table 1.

[1 mark]

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TABLE 1

Test Tube	Starch Concentration	Time Intervals (minutes)	Colour Changes
A 5 cm ³ starch solution		1	
		2	
		3	
		4	
		5	
A1 5 cm ³ distilled water		1	
		2	
		3	
		4	
		5	
B 4 cm ³ starch solution and 1 cm ³ distilled water		1	
		2	
		3	
		4	
		5	
B1 5 cm ³ distilled water		1	
		2	
		3	
		4	
		5	
C 3 cm ³ starch solution and 2 cm ³ distilled water		1	
		2	
		3	
		4	
		5	
C1 5 cm ³ distilled water		1	
		2	
		3	
		4	
		5	
D 2 cm ³ starch solution and 3 cm ³ distilled water		1	
		2	
		3	
		4	
		5	
D1 5 cm ³ distilled water		1	
		2	
		3	
		4	
		5	

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- (g) Based on the results recorded what can you deduce about the relationship between substrate concentration and enzyme activity?

(2 marks)

- (h) Comment on the purpose of the distilled water in test tubes A1 to D1.

[1 mark]

- (i) Give TWO reasons for the colour changes recorded over the five-minute test period, when Benedict's solution was added to the reaction mixtures.

[2 marks]

- (j) Outline the procedure that you might use to produce quantitative colour standards for the reaction mixtures using Benedict's solution.

[3 marks]

Total 16 marks

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2. (a) In a study to determine the effects of the steroidal plant hormone, 24-epibrassinolide (BL), on the mitotic index and growth of onion (*Allium cepa*) root tips, the effect of BL on cell division is measured by recording the number of cells in various stages of mitosis and the mitotic index (% of cells examined in mitosis) in controls and BL-treated root tips. The results are shown in Table 2.

Table 2: Effects of various concentrations of 24-epibrassinolide (BL) on the mitotic index of regenerating onion root tips. Data based on 400 cells counted from each of five root tips from each group (total of 2000 cells from each group).

Concentration of BL	# Cells in interphase	# Cells in prophase	# Cells in metaphase	# Cells in anaphase	# Cells telophase	# Cells counted	Mitotic index $\pm$ SD ¹
Control	1907	44	37	11	1	2000	4.65 ± 1.34
0.5 ppm	1918	52	15	12	3	2000	 ± 1.34
0.05 ppm	1869	82	28	20	1	2000	$6.55 \pm 0.69^{**}$
0.005 ppm	1807	112	46	24	11	2000	 $\pm 1.76^*$

¹Percent of cells in mitosis $\pm$ SD; *P $\leq$ 0.001; ** P $\leq$ 0.03.

Table taken from www.funpecrp.com.br/.../gmr0259_full_text.htm

- (i) Calculate the mitotic index (% cells examined in mitosis) for root tips treated with 0.5 ppm BL and 0.005 ppm BL.

0.5 ppm BL _____

0.005 ppm BL _____

[2 marks]

- (ii) Based on the results given in Table 2, what TWO pieces of information can be deduced about the effect of the steroidal plant hormone on cell division in onion root tips?

[2 marks]

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- (b) In experiments investigating the inheritance of mutations in *Drosophila melanogaster*, students are given a stock of mutant flies with the following mutations:

1. Mutant flies have lighter than normal body colour – **yellow body colour**
2. Mutant flies have wings which **lack the normal veins**

In the **first part of the experiment**, crosses were made as follows:

- A. True breeding normal (wildtype) virgin females are crossed with true breeding mutant males.
- B. True breeding mutant virgin females are crossed with normal (wildtype) true breeding males.

The results of cross A produced F_1 progeny (male and female flies) which were wildtype with normal body colour and wings with veins. For B, the male F_1 progeny had all the mutations, while none of the F_1 female offspring had mutations (all wildtype).

For the second part of the experiment, the F_1 progeny from the cross B, are crossed together. The results of the second cross are given in Table 3.

TABLE 3: RESULTS OF F_1 CROSSES FROM B

Phenotype	Total
All mutants	475
Wildtype	785
Normal body colour, mutant wings	78
Mutant body colour, normal wings	62
TOTAL	1400

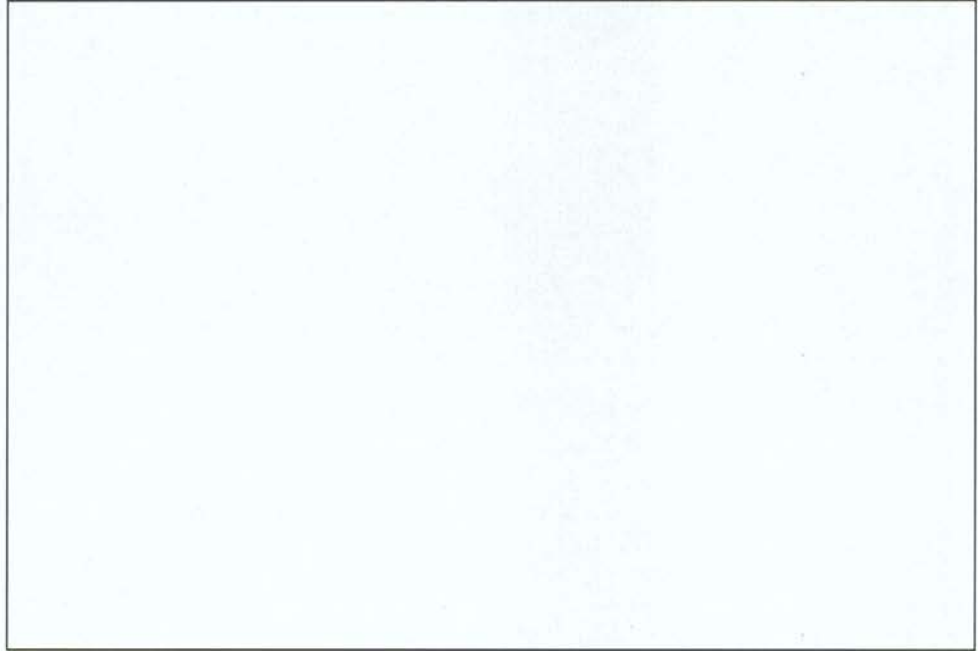
- (i) Based on the results of the **first part of the experiment**, what can be deduced about the nature of the inheritance of the mutations?

[1 mark]

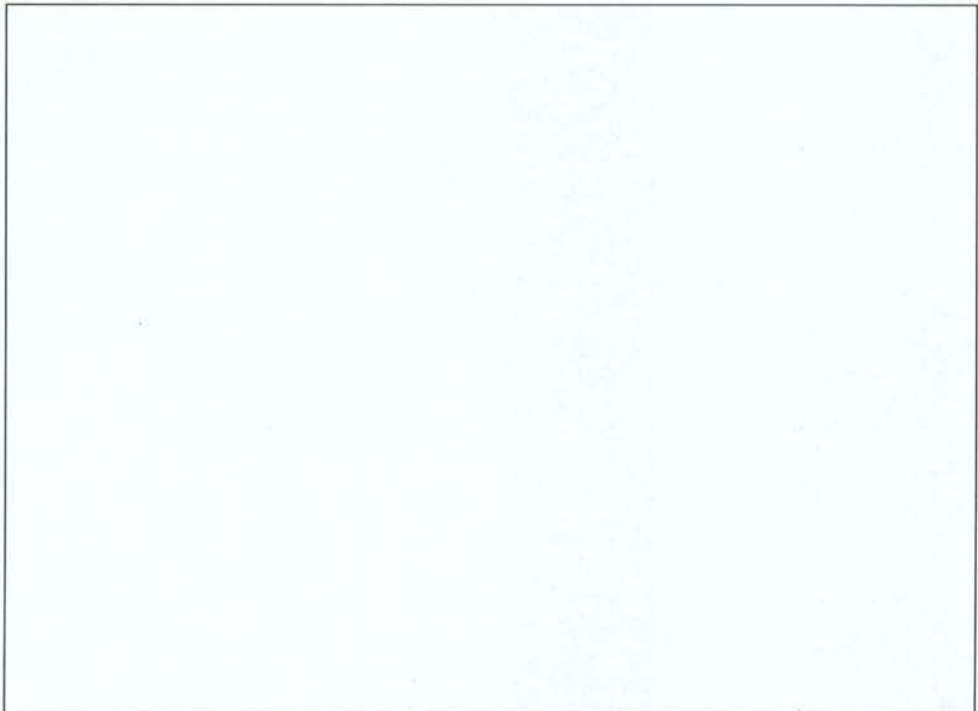
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- (ii) Draw a genetic diagram to show the genotypes and phenotypes of the parents and offspring of the two crosses, A and B, of the **first part of the experiment**.

CROSS A



CROSS B



[8 marks]

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- (iii) Using the data from Table 3 on page 6, calculate the percentage of offspring exhibiting non-parental phenotypes.

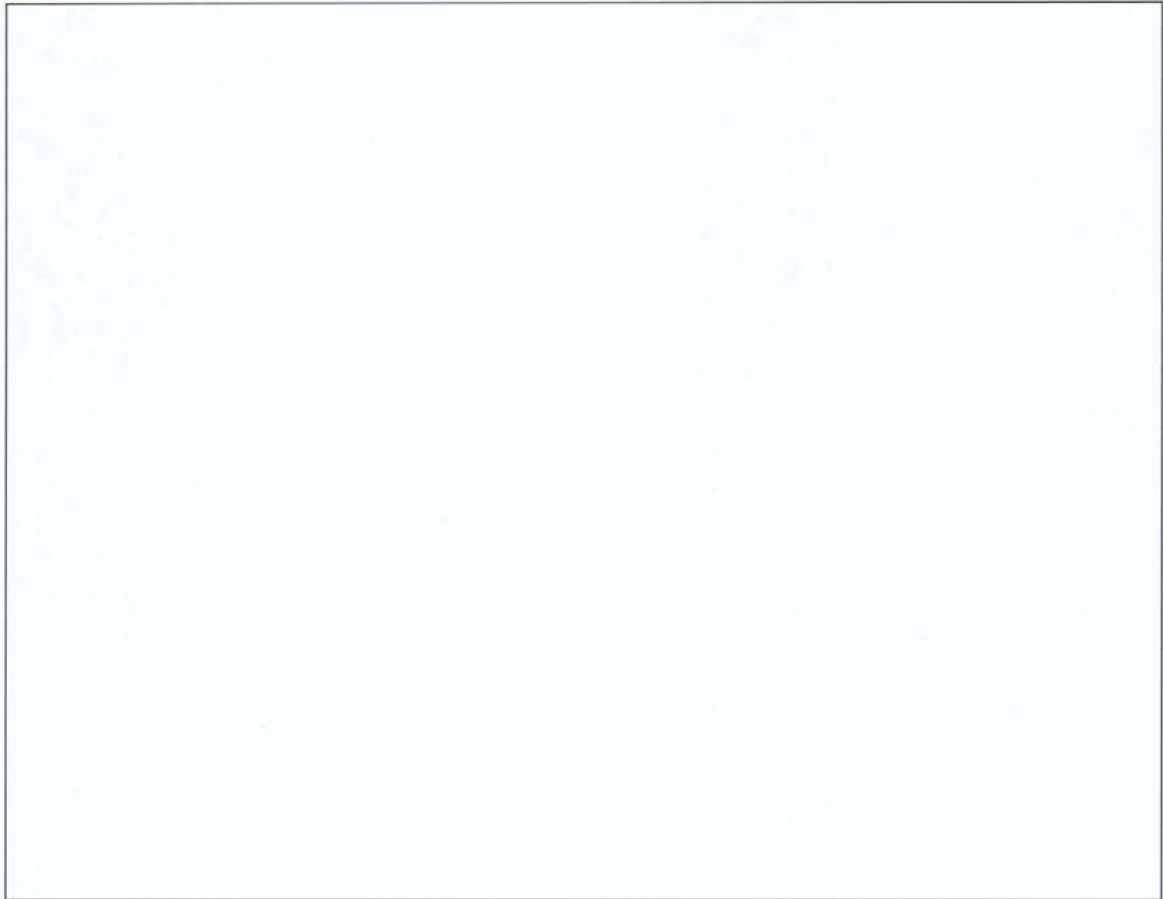
[2 marks]

- (iv) Explain the significance of finding offspring from the second cross (F_2) exhibiting non-parental conditions.

[1 mark]

Total 16 marks

3. (a) Specimen A is a stained transverse section of the ovary of a mammal.
- (i) Make a labelled plan drawing of Specimen A in the box below.



[6 marks]

- (ii) Examine the specimen and locate a mature Graafian follicle. Calculate the diameter of the mature oocyte. Show your calculations.

Calculations:

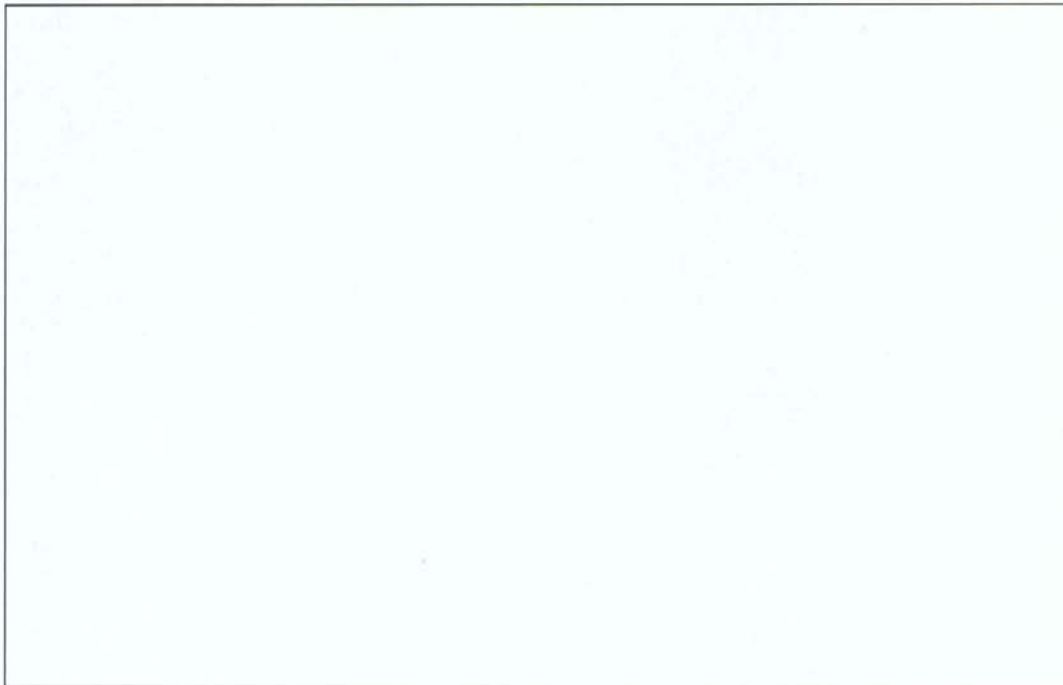
Diameter of oocyte _____

[2 marks]

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(b) Specimen B is a transverse section of an immature anther of a lily.

- (i) Examine the specimen and locate a pollen sac. Make a DETAILED labelled drawing of a pollen sac in the box below.



[6 marks]

- (ii) Comment on TWO **key** differences that you might expect to see if you were to observe the pollen sac in a section of a completely mature anther.

1. _____

2. _____

[2 marks]

Total 16 marks

END OF TEST

**DO NOT
WRITE ON
THIS PAGE**

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BIOLOGY

UNIT 01 – Paper 03/2

ALTERNATIVE TO INTERNAL ASSESSMENT

**INSTRUCTIONS FOR SETTING UP THE ALTERNATIVE TO
INTERNAL ASSESSMENT EXAMINATION**

Question 1.

For EACH candidate provide:

- One microscope
- Fungal amylase powder - 75g/l
- Corn starch solution - 1%
- Benedict's solution for reducing sugars
- Droppers
- Marble spot plates or white tiles
- Water bath at 37 °C
- Test tubes
- Test tube racks
- Pipettes - measuring
- Distilled water
- Small measuring cylinders capable of measuring 1 to 5 cm³ of solution
- Labels for the test tube

Question 2.

None

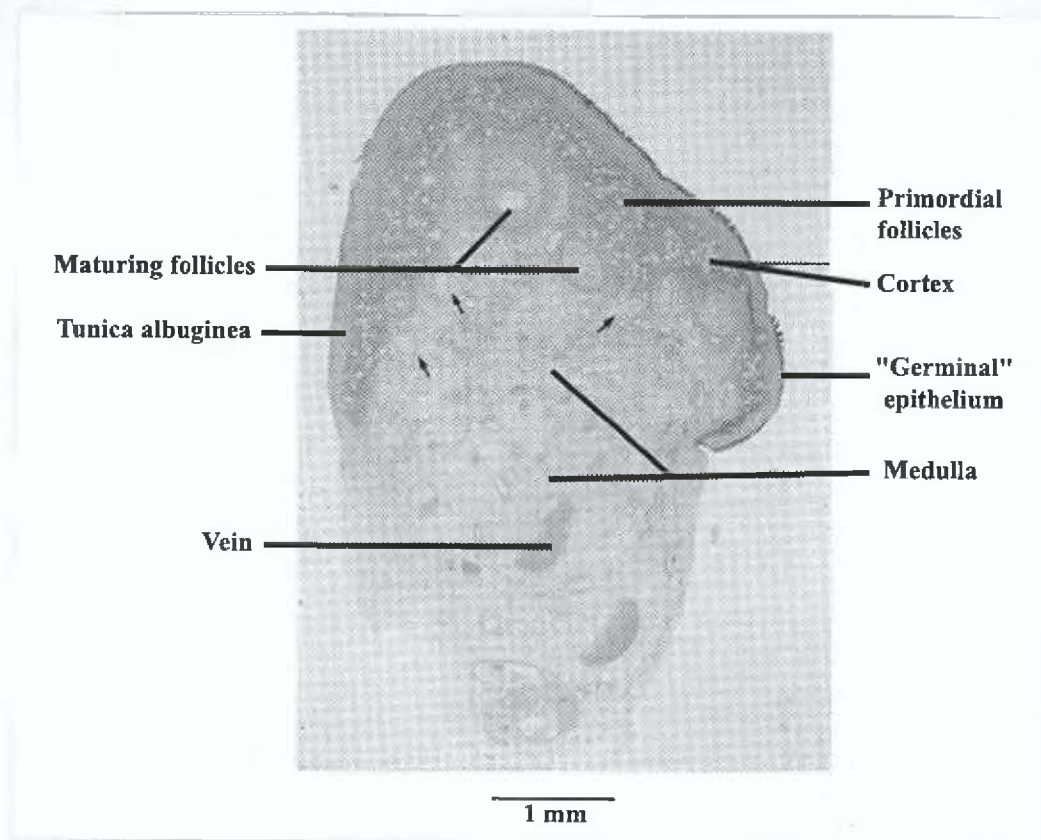
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Question 3.

For EACH candidate provide:

- Specimen A
- One compound microscope

Transverse section of the ovary of a mammal e.g. ovary of a monkey.



<http://www.anatomyatlases.org/MicroscopicAnatomy/Section13/Plate13245.shtml>

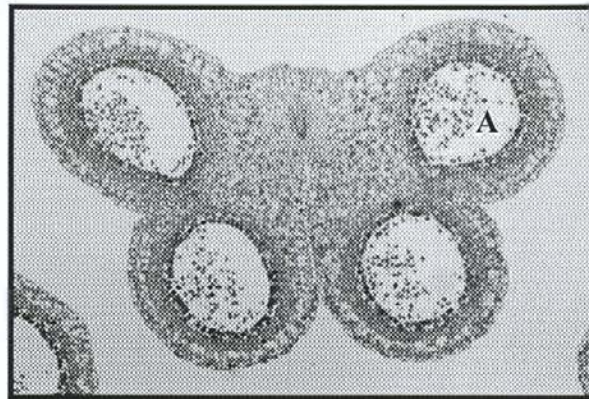
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For EACH candidate provide:

Specimen B

Transverse section of an immature anther of a lily

Young anther – Pollen sacs (A), with pollen grains developing inside.



www.funpecrp.com.br/.../gmr0259_full_text.htm

END OF INSTRUCTIONS

FORM TP 2009148 – SR



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MAY/JUNE 2009

SCHOOL/CENTRE NUMBER

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NAME OF SCHOOL/CENTRE

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CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

BIOLOGY

UNIT 01 – Paper 03/2

ALTERNATIVE TO INTERNAL ASSESSMENT

**SUPERVISOR'S REPORT ON THE ALTERNATIVE TO
INTERNAL ASSESSMENT EXAMINATION**

When completed, this report **MUST** accompany
candidates' answer booklets for the Practical Examination.

SUPERVISOR'S NAME

(Please print)

SUPERVISOR'S SIGNATURE

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02107032/CAPE/2009 – SR

The attention of supervisors is drawn to the following:

(a) ASSISTANCE TO CANDIDATES

Candidates **MUST** be told before the test begins that:

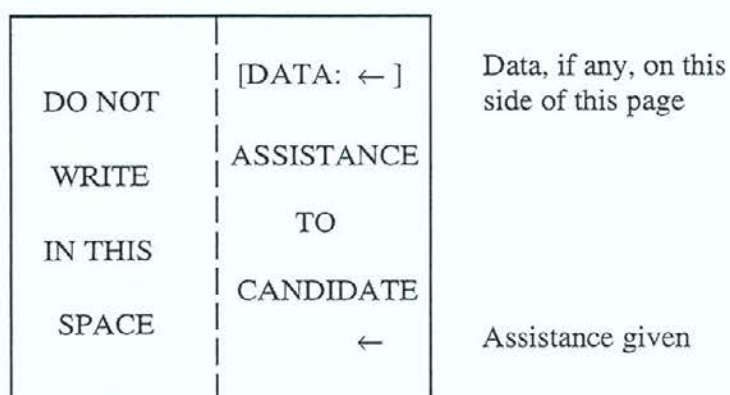
- (i) Help will be available to those candidates who need it in order to proceed with the experiment, but that such help will be given to a candidate only if he/she requests it. Assistance should be confined to guidance in setting up apparatus or describing procedure to enable candidates to obtain readings. Help with the interpretation of graphs, the determination of gradients or calculations and formulae is **NOT** to be given.
- (ii) Wherever such assistance is given, a full report must be submitted to the examiners of the nature and extent of it.
- (iii) Candidates who receive help are liable to suffer a loss of marks.

Supervisors will appreciate the need for such reports to be accurate if candidates who receive help are to be fairly assessed. They are therefore requested in submitting their reports to state the nature and extent of the assistance given in the space allotted in the candidate's answer booklet (see diagram in (b) below), and also in the space provided on page 5 in this report form. In any report it is important to include the candidate's number as well as name.

N.B. Help is not to be offered to any candidate.

(b) DATA TO BE SUPPLIED TO CXC

Data are to be supplied on this report form in the space provided. In addition, they should be given on the right-hand half of the inside of the front cover of each candidate's answer booklet. (See diagram below.) This should be done only **AFTER THE COMPLETION OF THE EXAMINATION**.



FRONT INSIDE COVER OF ANSWER BOOKLET

- (c) Only apparatus stipulated in the "Instructions to Supervisors" booklet is to be supplied to the candidate. The issue of additional equipment or apparatus is **NOT** allowed.
- (d) The declaration form on page 6 is to be signed as indicated before this report is submitted.

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Information required from Supervisor:

Question 1.

Source of amylase

Concentration of amylase

Source of starch

Concentration of starch

Question 2.

None

Question 3.

Type of mammalian ovary

Type of anther

GO ON TO THE NEXT PAGE

General Remarks. Here the Supervisor may give an account of any incident(s) which he/she thinks should be communicated to the examiners, in order that they may fairly assess the performance of the candidate and the quality of his/her work.

GO ON TO THE NEXT PAGE

Report on Assistance Given. (If necessary, continue on page 6.)

GO ON TO THE NEXT PAGE

Declaration: (Delete as appropriate).

I/We declare that NO assistance was given to any candidate during the examination.

I/We declare that ALL assistance given during the examination has been reported above.

Teacher's Signature

Supervisor's Signature

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Date: